

# FEDERICO BRUZZONE • CURRICULUM VITAE

## PERSONAL INFORMATION

Born in Magenta (MI), Italy on 7th of **March 2000**  
Resident of Via F. Turati 75/F, Arluno (MI), 20004  
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## CONTACT INFORMATION

[github.com/FedericoBruzzone](https://github.com/FedericoBruzzone)  
[@federicobruzzone](https://twitter.com/federicobruzzone)  
[in/federico-bruzzone](https://www.linkedin.com/in/federico-bruzzone)  
[@fedebruzzone7](https://x.com/fedebruzzone7)  
[u/FedericoBruzzone](https://www.youtube.com/u/FedericoBruzzone)

PhD Candidate in Computer Science. Programming Languages and Compilers enthusiast. Also, a Sound Engineer and Music Composer. For more information, visit my personal [website](#).

## SCIENTIFIC PUBLICATIONS

- [1] F. Bruzzone, W. Cazzola, M. Brancaleoni, and D. Pellegrino, “Sink or SWIM: Tackling Real-Time ASR at Scale,” *IEEE Transactions on Audio, Speech and Language Processing*, no. , pp. 1–11, 2026, doi: [10.1109/TASLPRO.2026.3675780](https://doi.org/10.1109/TASLPRO.2026.3675780). **Journal Ranked Q1 on Scimago** — [bib] [pdf] [IEEE] [arXiv]  
[2] F. Bruzzone, W. Cazzola, and L. Favalli, “Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families,” *Journal of Systems and Software*, p. 112554, Sept. 2025, doi: [10.1016/j.jss.2025.112554](https://doi.org/10.1016/j.jss.2025.112554). **Journal Ranked Q1 on Scimago** — [bib] [pdf] [SpringerLink] [arXiv]

## PREPRINTS PUBLICATIONS

- [3] F. Bruzzone, W. Cazzola, and L. Favalli, “Generalized Software Product Line Extraction.” [Online]. Available: <https://arxiv.org/abs/2605.28989> — [bib] [pdf] [arXiv]  
[4] F. Bruzzone and W. Cazzola, “Meta-Monomorphizing Specializations.” [Online]. Available: <https://arxiv.org/abs/2602.12973> — [bib] [pdf] [arXiv]  
[5] F. Bruzzone, W. Cazzola, and L. Favalli, “From Separate Compilation to Sound Language Composition.” [Online]. Available: <https://arxiv.org/abs/2602.03777> — [bib] [pdf] [arXiv]  
[6] F. Bruzzone, W. Cazzola, M. Brancaleoni, and D. Pellegrino, “Sink or SWIM: Tackling Real-Time ASR at Scale.” [Online]. Available: <https://arxiv.org/abs/2601.17097> — [bib] [pdf] [arXiv] **[PUBLISHED]**  
[7] F. Bruzzone, W. Cazzola, and L. Favini, “Prioritizing Configuration Relevance via Compiler-Based Refined Feature Ranking.” [Online]. Available: <https://arxiv.org/abs/2601.16008> — [bib] [pdf] [arXiv]  
[8] F. Bruzzone, W. Cazzola, and L. Favalli, “Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families,” *Journal of Systems and Software*, p. 112554, Sept. 2025, doi: [10.1016/j.jss.2025.112554](https://doi.org/10.1016/j.jss.2025.112554). — [bib] [pdf] [arXiv] **[PUBLISHED]**

## EDUCATION

|                     |                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2024-Present</b> | <b>PhD Candidate</b> in Computer Science at the <a href="#">ADAPT Lab</a> , University of Milan<br>Under the supervision of <i>W. Cazzola</i> , my research focuses <b>compiler/IR construction</b> and <b>programming languages, analysis and transformation</b> of <b>optimizing compilers</b> as well as <b>type systems</b> and <b>support tools</b> (e.g., LSP). |
| <b>2022-2024</b>    | MSc in Computer Science at University of Milan ( <i>110/110 cum laude</i> , <i>W. Cazzola</i> , 15/07/2024)<br>Thesis: “ <b>Toward a Modular Approach for Type Systems and LSP Generation</b> ” .                                                                                                                                                                     |
| <b>2019-2022</b>    | BSc in Musical Computer Science at University of Milan (13/10/2022)                                                                                                                                                                                                                                                                                                   |
| <b>2011-2019</b>    | Piano and Music Composition at I.S.S.M. Novara Conservatory                                                                                                                                                                                                                                                                                                           |
| <b>2014-2019</b>    | Diploma in Computer Science and Telecommunications at E. Alessandrini                                                                                                                                                                                                                                                                                                 |

## RESEARCH ACTIVITIES

|                     |                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2024-Present</b> | <b>MUSEMI Session Chair, Speaker and Co-organizer</b>                                                                                      |
| <b>May 2026</b>     | <b>Committee Member</b> at the Int. Conf. on Software Language Engineering ( <a href="#">SLE 2026</a> ), <i>ACM</i> , <b>B on CORE</b>     |
| <b>Nov 2025</b>     | <b>Reviewer</b> for the Journal of Computer Languages ( <a href="#">COLA</a> ), <i>Elsevier</i> , <b>C on CORE</b>                         |
| <b>Jun 2025</b>     | <b>Reviewer</b> for the Journal of Software and Systems Modeling ( <a href="#">SoSyM</a> ), <i>Springer</i> , <b>Q1 on Scimago</b>         |
| <b>Dec 2025</b>     | <b>Reviewer</b> for the Eur. Conf. on Object Oriented Programming ( <a href="#">ECOOP 2026</a> ), <i>ACM</i> , <b>A on CORE</b>            |
| <b>2-6 Jun 2025</b> | <b>Participant</b> at <Programming> 2025 conference                                                                                        |
| <b>Apr-Jun 2025</b> | <b>Committee Member</b> at the Int. Conf. on Software Language Engineering ( <a href="#">SLE 2025</a> ), <i>ACM</i> , <b>B on CORE</b>     |
| <b>Mar 2025</b>     | <b>Reviewer</b> for the Journal of Systems and Software ( <a href="#">JSS</a> ), <i>Elsevier</i> , <b>Q1 on Scimago</b>                    |
| <b>2-7 Sep 2024</b> | <b>Student Volunteer</b> at the Int. Conf. on Functional Programming ( <a href="#">ICFP 2024</a> )                                         |
| <b>2022-2024</b>    | Research <b>Internship</b> at <a href="#">ADAPT Lab</a> , working on modular type systems and LSP generation for <a href="#">Neverlang</a> |

**THESIS SUPERVISION**

|            |                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 23/02/2026 | D. Cerato, <i>Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization</i> , BSc, <b>103</b> |
| 10/04/2025 | D. Pellegrino, <i>Scalable Multi-client Real-time Whisper</i> , BSc, <b>96</b>                                                                    |
| 09/04/2025 | A. Longoni, <i>GUIDE: Graphical User Interface Development Environment</i> , MSc, <b>110L</b>                                                     |
| 09/04/2025 | L. Albani, <i>New Generalized Protocol For Software Product Line Extraction And Configuration</i> , MSc, <b>110L</b>                              |
| 09/04/2025 | G. Esposito, <i>Fr3D: A Framework for DAP-compatible DSL-oriented Debugging</i> , MSc, <b>110L</b>                                                |
| 24/02/2025 | L. Favini, <i>RustyEx: Instrumenting rustc to Extract Feature Dependency Graphs</i> , BSc, <b>102</b>                                             |

**GRADUATE COURSES**

|           |                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------|
| 2025-2026 | Mathematical Logic (Art. 45), <b>BSc</b> in CS, University of Milan, <i>S. Aguzzoli</i>                  |
| 2024-2025 | Programming 1 (Art. 45), <b>BSc</b> in CS, University of Milan, <i>L. Capra</i> (coordinator W. Cazzola) |
| 2024-2025 | Mathematical Logic (Art. 45), <b>BSc</b> in CS, University of Milan, <i>S. Aguzzoli</i>                  |
| 2023-2024 | Mathematical Logic, <b>BSc</b> in CS, University of Milan, <i>S. Aguzzoli</i>                            |
| 2023-2024 | Computer Science, <b>BSc</b> in Comm. and Society, University of Milan, <i>A. Momigliano</i>             |
| 2023-2024 | Programmin 1, <b>BSc</b> in CS, University of Milan, <i>A. Trentini</i> (coordinator P. Boldi)           |
| 2023-2024 | Programming in Python, <b>MSc</b> in Chemistry, University of Milan, <i>M. Monga</i>                     |

**ADDITIONAL ACTIVITIES**

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| 2023-Present | <b>Private Tutoring</b> in Computer Science, Mathematics and Physics                      |
| Nov 2024     | <b>Collaborator</b> at the <a href="#">Bebras Challenge</a> , <a href="#">ALaDDIn Lab</a> |
| Jan-Jun 2024 | <b>Organizer</b> of the BSc Computer Science Laboratories, <a href="#">ALaDDIn Lab</a>    |
| Jan-Jun 2024 | <b>Collaborator</b> of the Workshops for schools, <a href="#">ALaDDIn Lab</a>             |
| Nov 2023     | <b>Collaborator</b> at the <a href="#">Bebras Challenge</a> , <a href="#">ALaDDIn Lab</a> |

**PUBLIC CONTRIBUTIONS (SELECTED)**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-Present    | Maintainer of the <a href="#">LLVM Pass Template</a> , a C++ template project to quickly create new LLVM passes. This project allows developers to quickly bootstrap, test, and benchmark new out-of-tree LLVM passes with minimal effort — that is, without the need to build LLVM from sources.                                                                                                                                         |
| 2026/06/12      | <a href="#">LLVM (MLIR)</a> : extend createReadOrMaskedRead/createWriteOrMaskedWrite with permutation map support<br>Extends the createReadOrMaskedRead and createWriteOrMaskedWrite utilities in VectorUtils with permutation map support. The affine super-vectorizer is updated to use these utilities instead of constructing TransferReadOp/TransferWriteOp directly.                                                                |
| 2026/06/09      | <a href="#">LLVM (MLIR)</a> : affine-super-vectorize now emits in_bounds transfer ops<br>Fixes affine-super-vectorize to emit vector.transfer_read/write with the in_bounds attribute when statically provable, avoiding unnecessary masked intrinsics overhead on AArch64/NEON.                                                                                                                                                          |
| 2025-Present    | Maintainer of the <a href="#">Papers on Compiler Optimizations: Analysis and Transformations</a> , a curated list of scientific publications on compiler optimizations and related topics.<br>This repository curates a chronologically sorted list of influential papers on compiler optimization, from the seminal works of 1952 through the advanced techniques of 1994.                                                               |
| 2025/09-Present | <b>Maintainer</b> of <a href="#">scribe</a> , a minimalist, opinionated LaTeX document class for academic technical writing and presentations.<br>Scribe provides a clean, professional layout with callout boxes, theorem environments, line numbering, and a matching Beamer theme.                                                                                                                                                     |
| 2025-Present    | <b>Maintainer</b> of the <a href="#">Tide Compiler</a> , an backend-agnostic IR and compiler framework written in <b>Rust</b> .<br><b>Tide</b> aims to be a modular and extensible framework for building compilers and language tools, prioritizing simplicity and ease of use. From its quasi-SSA IR to its flexible backend architecture, Tide provides a solid foundation for developing compilers for various programming languages. |
| 2025-Present    | Contributions to the <a href="#">Rustworkx</a> graph library, focusing on implementing the Closeness Centrality algorithm for weighted graphs (following Newman's method) and integrating it into the Python bindings.<br>• <a href="#">Generalizing Closeness centrality to weighted networks using Newman method</a> (issue 1384)                                                                                                       |
| 2024-Present    | <b>Maintainer</b> of the cross-platform <a href="#">tgt</a> project and <a href="#">tclib-rs</a> TDLib bindings written in <b>Rust</b> .<br><a href="#">Tgt</a> is a TUI (Terminal User Interface) client for Telegram, built using the <a href="#">tclib-rs</a> library, which provides safe and idiomatic Rust bindings to the official TDLib (Telegram Database Library) C++ library. Thanks to CI/CD                                  |

pipelines, we ensure (i) that the projects build and work correctly on Linux, macOS, and Windows, (ii) automatic releases on GitHub and crates.io, and (iii) automatic documentation generation.

## TECHNICAL SKILLS

|                          |                                                                                                                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>         | <b>Rust, Python, C/C++, Go</b> , OCaml, Java, Scala, Kotlin, Erlang, Lua, Dart, PHP, HTML/CSS, SQL, Bash, TeX                                                                                                                                                                              |
| <b>Systems/Tooling</b>   | <b>Git, CI/CD, Docker, GDB, Valgrind</b> , Build Systems (e.g., CMake, Cargo, Pip), Cross-language <b>Linking</b> , Static/Dynamic Libraries, and <b>FFI</b> (e.g., C bindgen)                                                                                                             |
| <b>Area of Expertise</b> | <b>Compiler Construction</b> and <b>Optimizations</b> , Programming Languages, <b>IR Design</b> and <b>Implementation</b> , Type Systems, Language Support Tools (e.g., LSP), Static Analysis, Parsing Techniques and Parser Generators (e.g., ANTLR), <b>Rustc</b> Internals, <b>LLVM</b> |

## MUSICAL ACTIVITIES

|                     |                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>2021-Present</b> | <b>Sound Engineer and Music Producer</b> , in collaboration with <a href="#">Believe</a> and as a <a href="#">SIAE</a> member. |
| <b>2006-Present</b> | <b>Pianist and Music Composer</b> , composing original pieces and arrangements for piano and various ensembles                 |
| <b>2019-Present</b> | <b>Piano and Music Teacher</b> , teaching piano and music theory and composition to students of all ages and levels            |

## GRANTS AND FELLOWSHIPS

|                  |                                                                                         |
|------------------|-----------------------------------------------------------------------------------------|
| <b>2024</b>      | <b>56th Top Github Public Contributor</b> in Italy out of 958                           |
| <b>2023-2024</b> | Scholarship for the MSc in Computer Science, awarded by the University of Milan         |
| <b>2020-2024</b> | Scholarship for the BSc in Musical Computer Science, awarded by the University of Milan |

## TONGUES

|                |                                             |
|----------------|---------------------------------------------|
| <b>Italian</b> | Mother tongue                               |
| <b>English</b> | Level CEFR B2 (SLAM at University of Milan) |
| <b>Spanish</b> | Base (A1-A2)                                |

Milan, 12/06/2026